

Daniel Cohen

Mathematics & Statistics Professional | Quantitative Analytics | AI Decision-Support Applications

GitHub: github.com/Coakley11 | LinkedIn: available via portfolio contact

Portfolio: <https://coakley11.github.io/master-portfolio-hub/>

PROFESSIONAL SUMMARY

Mathematics and Statistics professional with an M.A. in Statistics & Applied Mathematics (GPA 3.93), MBA in Finance & Investments (Finance GPA 4.00), actuarial exam foundation (SOA P, FM, MFE), and a deployed seven-app analytics portfolio. Builds decision-support products in Python/Streamlit spanning portfolio risk, sports analytics, simulation labs, and AI evaluation - with SQL/Excel reporting fluency and end-to-end ownership from data through deployment.

EDUCATION

M.A. Statistics & Applied Mathematics - Hunter College

GPA: 3.93

Overall GPA: 3.88 | Finance GPA: 4.00

Magna Cum Laude | Arthur D. Gayer Memorial Award in Economics

MBA Finance

B.A. Mathematics

CREDENTIALS AND CERTIFICATIONS

- Society of Actuaries - Exam P/1 (Probability), FM/2 (Financial Mathematics), MFE (Models for Financial Economics)
- NYS Certification - CST Mathematics (004), Educating All Students (EAS)

EXPERIENCE

Analytics & AI Portfolio Developer (2024 - Present)

Independent - Daniel AI Suite

- Designed, built, and deployed seven interconnected Streamlit applications for portfolio analytics, fantasy baseball operations, applied math decision labs, playoff intelligence, music practice tooling, AI transition scenarios, and suite orchestration
- Implemented decision scoring, portfolio health diagnostics, Monte Carlo and efficient frontier modeling, live-data fallbacks, and cross-app resume/insight workflows with Supabase-backed persistence patterns
- Authored SQL & Excel analytics workbooks with KPI dashboards, pivot exercises, and 42+ structured queries including AI evaluator practice datasets

Full-Time High School Mathematics Teacher (2024 - Present)

The Montfort Academy

- Instruct Algebra II, Geometry, and Pre-Calculus with emphasis on quantitative reasoning, functions, and data interpretation
- Support Chemistry and Physics laboratory sessions requiring measurement, precision, and analytical documentation
- Apply structured problem-solving frameworks that parallel analytics debugging and hypothesis-driven analysis

Permanent Full-Time Substitute Teacher (2021 - 2024)

Yeshiva Har Torah

- Delivered K-8 secular and Judaic studies instruction across multiple subjects and grade levels
- Prepared science laboratory materials and supported hands-on quantitative observation activities
- Maintained classroom data tracking for student progress and instructional adjustment

Mathematics Instructor (2020)

Pace University

- Taught Mathematics for Life, College Algebra, and Calculus I - connecting algebraic reasoning to applied decision contexts
- Emphasized interpretation of quantitative results, not just procedural computation

Mathematics Instructor (2020)

Yeshiva University

- Led Calculus III Problem Solving Seminar focused on multi-step analytical reasoning and proof-style logic

Full-Time Lecturer / Adjunct Lecturer (2014 - 2019)

Kingsborough Community College (CUNY)

- Taught Quantitative Reasoning, College Algebra, Intermediate Algebra, Principles of Mathematics, and Pre-Algebra
- Developed curriculum reinforcing probability intuition, algebraic modeling, and statistical literacy at scale
- Supported diverse learners in translating word problems into structured quantitative frameworks

Adjunct Lecturer (2015 - 2019)

Queens College (CUNY)

- Instructed Probability & Statistics, Pre-Calculus, Calculus, College Algebra, and Math for Elementary School Teachers
- Taught hypothesis-testing concepts, probability distributions, and regression intuition at undergraduate level

Business Statistics Lab Professor (2012)

Queens College

- Led Business Statistics labs with Excel instruction, computer-lab analytics exercises, and data interpretation assignments
- Connected descriptive statistics, inference, and spreadsheet modeling to business decision scenarios

College Assistant & Academic Tutor (2011 - 2014)

Queens College

- Tutored Statistics, Economics, and Finance with focus on quantitative support and exam preparation
- Diagnosed knowledge gaps and built targeted practice for data interpretation and financial quantitative reasoning

Mathematics & Statistics Tutor (2011 - Present (periodic))

Tutor Doctor | Club Z | Private | SAT Prep

- Provided individualized instruction in Mathematics, Economics, Statistics, and Finance from remedial through advanced undergraduate levels
- Delivered SAT mathematics preparation emphasizing problem decomposition and analytical speed

SELECTED PROJECTS

Baseball Analytics | Python | Streamlit | scikit-learn

- Built full-season fantasy platform with Decision Score drafts, live draft rooms, shared leagues, and lineup/trade workflows
- Integrated research tools and AML insight handoff for analytical follow-up questions

Investment Explorer | Python | Streamlit | yfinance

- Shipped live portfolio health scoring, Monte Carlo, efficient frontier, and ETF overlap diagnostics
- Designed beginner/advanced modes on a shared analytics core; deployed on Streamlit Cloud

Applied Mathematical Intelligence | Python | Streamlit

- Built modular decision labs for EV, prediction, disease modeling, and AI training interpretation
- Routed analytical questions across the suite via Command Center insight handoffs

SQL & Excel Analytics Workbooks | Excel | SQL | pandas

- Created AI evaluator, investment, quant, and credit-risk workbooks with KPI dashboards and SQL practice
- Demonstrates fluency across Python products and spreadsheet-based business reporting

TECHNICAL SKILLS

Statistics | Probability | Hypothesis Testing | Python | SQL | Excel | Pivot Tables | Pandas | Streamlit | Data Visualization | Machine Learning | Monte Carlo Simulation | Portfolio Optimization | Financial Analysis | Dashboard Development | AI Evaluation | Product Analytics | Git / GitHub | AI-Assisted Development | Decision Support Systems

TARGET ROLES

Data Analyst | Product Analyst | AI Evaluator | AI Trainer | Business Intelligence Analyst | Research Analyst | Junior Data Scientist | Quantitative Analyst | Financial Analyst